

BMS & Inverter Functional Safety Validation Test Report

HIL & Dynamometer Empirical Verification per ISO 26262 ASIL-D

1. High-Voltage Battery Protection Tests

- Test Case TC-BMS-001 (Overvoltage): Overvoltage injected at 4.255 V; persistent for 100 ms triggered contactor trip in 14.2 ms. Verdict: Pass
- Test Case TC-BMS-002 (Pack Voltage Span): Validated continuous pack monitoring from 380.0 V DC to 820.0 V DC across full operating range. Verdict: Pass
- Test Case TC-BMS-003 (Current Sensor Dynamic Range): Current sensor evaluated at discrete discharge test points of +100.0 A, +200.0 A, and -100.0 A. Verdict: Pass
- Test Case TC-BMS-004 (Balancing Thermal Limits): Thermal dissipation measurement on balancing circuitry: balancing current clamp accuracy within 2%. Verdict: Pass

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2. Battery Monitoring Diagnostics & Isolation Records

- Section 4.1: Isolation monitoring circuit layout reviewed. Isolation fault response timing test scheduled for Phase 3 validation.

3. Traction Inverter Safety Shutdown & Discharge Tests

- Test Case TC-INV-011 (Phase Overcurrent): Short-circuit test with DESAT protection: peak current 680 A detected and all gates disabled
- Test Case TC-INV-012 (Active DC-Link Discharge): Active discharge test: DC-link voltage decayed from 800 V to 42.5 V in 3.4 seconds

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4. Inverter Dynamometer Mapping & Sensor Bandwidth

- Test Case TC-INV-013 (Efficiency Profile): Dynamometer efficiency test conducted at 4000 rpm from 100 Nm to 250 Nm showed efficiency
- Test Case TC-INV-014 (Resolver Angle Tracking): Resolver tracking error measured at +25°C ambient up to 15000 rpm was ± 0.14 deg
- Test Case TC-INV-016 (Current Hall Bandwidth): Phase current sensor frequency response measured -3dB cutoff at 160.0 kHz. Bandwidth

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5. Microcontroller Watchdog & Brownout Reset Tests

- Test Case TC-SAF-071 (Window Watchdog): Watchdog timing verification: SPI challenge-response verified at 15.2 ms interval. Forced
- Test Case TC-SAF-072 (Core Brownout): Brownout injection test: Vcore reduced to 2.94 V triggered PMIC hard RESET in 2.1 μs. Ver
- Test Case TC-SAF-073 (Lockstep Core Error Injection): Lockstep error injection on Core 0 ALU registers triggered safety alarm in 1 cl

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6. Safety Cross-Check & Fault Reaction Times

- Test Case TC-SAF-074 (ADC Redundant Cross-Check): Cross-check algorithm verified at room temperature with 2.0% injected delta,
- Test Case TC-SAF-075 (Safe Torque Off FTTI): Fault injection test: measured total elapsed time from overcurrent trigger to STO state
- Test Case TC-SAF-076 (Flash Memory ECC Handling): Memory fault injection: double-bit flash error triggered standard exception han

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7. Functional Safety FMEDA & Cybersecurity TARA Calculations

- Quantitative Safety Metrics: FMEDA quantitative calculation report indicates theoretical SPFM of 99.4% and LFM of 91.2%.
- Cybersecurity Risk Assessment: TARA analysis report section 4 evaluates 42 threat vectors and assigns theoretical CAL ratings.